

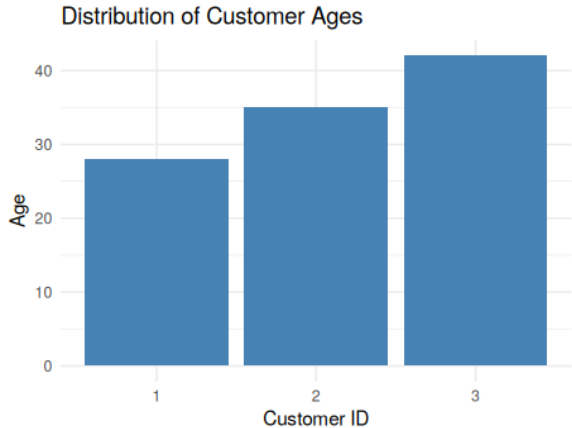
Lab Experiments 16-20

Python and R: Code and Output (Side by Side)

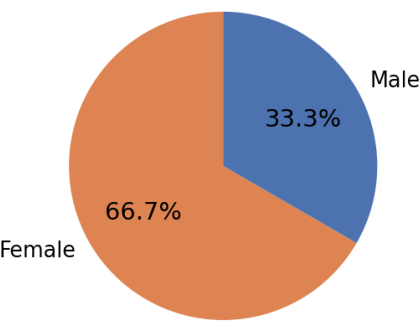
Date: 28/07/2026

Experiment 16

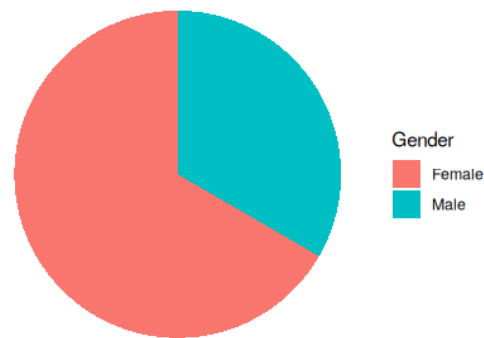
Customer Demographics Analysis

Python	R																
<pre>import matplotlib.pyplot as plt import pandas as pd df = pd.DataFrame({ 'CustomerID': [1, 2, 3], 'Age': [28, 35, 42], 'Gender': ['Female', 'Male', 'Female'], 'Income': [50000, 60000, 75000] }) # 1. Bar chart - distribution of customer ages plt.figure(figsize=(6, 5)) plt.bar(df['CustomerID'].astype(str), df['Age'], color='steelblue') plt.title("Distribution of Customer Ages") plt.xlabel("Customer ID"); plt.ylabel("Age") plt.tight_layout() plt.show() # 2. Pie chart - distribution by gender gender_counts = df['Gender'].value_counts() plt.figure(figsize=(6, 6)) plt.pie(gender_counts.values, labels=gender_counts.index, autopct='%1.1f%%', startangle=90, colors=['#DD8452', '#4C72B0']) plt.title("Customer Distribution by Gender") plt.tight_layout() plt.show() # 3. Table - demographic information per customer fig, ax = plt.subplots(figsize=(7, 2)) ax.axis('off') tbl = ax.table(cellText=df.values, colLabels=df.columns, loc='center', cellLoc='center') tbl.auto_set_font_size(False) tbl.set_fontsize(11) tbl.scale(1, 1.8) plt.title("Customer Demographic Data", pad=20) plt.tight_layout() plt.show()</pre>	<pre>library(ggplot2) library(gridExtra) library(grid) df <- data.frame(CustomerID = 1:3, Age = c(28,35,42), Gender = c('Female','Male','Female'), Income = c(50000,60000,75000)) # 1. Bar chart - distribution of customer ages p1 <- ggplot(df, aes(x=factor(CustomerID), y=Age)) + geom_col(fill="steelblue") + labs(title="Distribution of Customer Ages", x="Customer ID", y="Age") + theme_minimal() print(p1) # 2. Pie chart - distribution by gender gender_counts <- as.data.frame(table(df\$Gender)) colnames(gender_counts) <- c('Gender', 'Count') p2 <- ggplot(gender_counts, aes(x="", y=Count, fill=Gender)) + geom_col(width=1) + coord_polar("y") + labs(title="Customer Distribution by Gender") + theme_void() print(p2) # 3. Table - demographic information per customer grid.newpage() grid.table(df, rows=NULL)</pre>																
Output	Output																
 A bar chart titled "Distribution of Customer Ages". The x-axis is labeled "Customer ID" and has three categories: 1, 2, and 3. The y-axis is labeled "Age" and ranges from 0 to 40. There are three blue bars representing the ages of the three customers: Customer 1 is 28, Customer 2 is 35, and Customer 3 is 42. <table border="1"><thead><tr><th>Customer ID</th><th>Age</th></tr></thead><tbody><tr><td>1</td><td>28</td></tr><tr><td>2</td><td>35</td></tr><tr><td>3</td><td>42</td></tr></tbody></table>	Customer ID	Age	1	28	2	35	3	42	 A bar chart titled "Distribution of Customer Ages". The x-axis is labeled "Customer ID" and has three categories: 1, 2, and 3. The y-axis is labeled "Age" and ranges from 0 to 40. There are three blue bars representing the ages of the three customers: Customer 1 is 28, Customer 2 is 35, and Customer 3 is 42. <table border="1"><thead><tr><th>Customer ID</th><th>Age</th></tr></thead><tbody><tr><td>1</td><td>28</td></tr><tr><td>2</td><td>35</td></tr><tr><td>3</td><td>42</td></tr></tbody></table>	Customer ID	Age	1	28	2	35	3	42
Customer ID	Age																
1	28																
2	35																
3	42																
Customer ID	Age																
1	28																
2	35																
3	42																

Customer Distribution by Gender



Customer Distribution by Gender



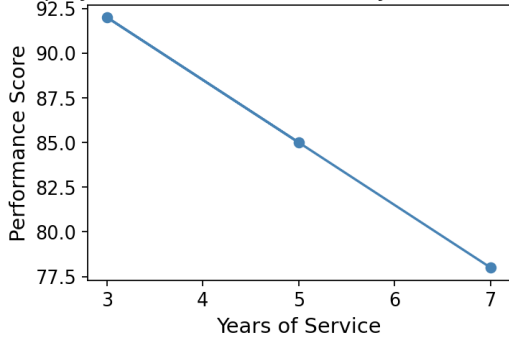
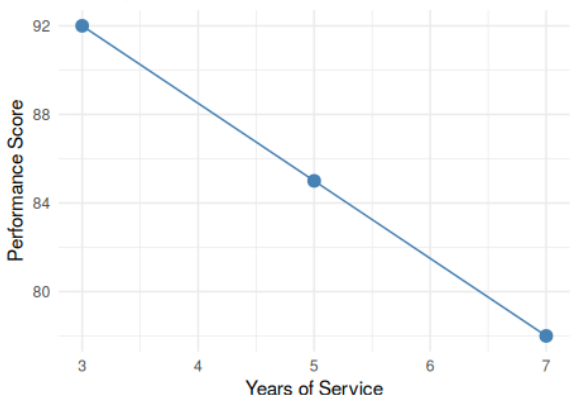
Customer Demographic Data

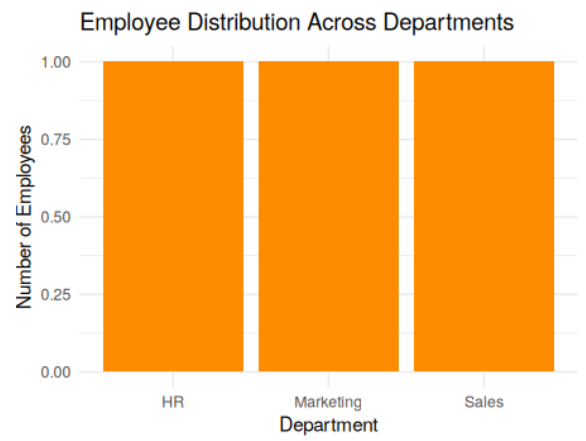
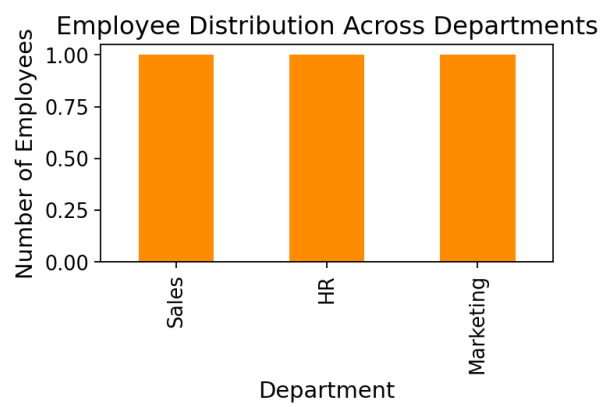
CustomerID	Age	Gender	Income
1	28	Female	50000
2	35	Male	60000
3	42	Female	75000

CustomerID	Age	Gender	Income
1	28	Female	50000
2	35	Male	60000
3	42	Female	75000

Experiment 17

Employee Performance Analysis

Python	R																
<pre>import matplotlib.pyplot as plt import pandas as pd df = pd.DataFrame({ 'EmployeeID': [1, 2, 3], 'Department': ['Sales', 'HR', 'Marketing'], 'YearsOfService': [5, 3, 7], 'PerformanceScore': [85, 92, 78] }) # 1. Line chart - performance trend over time # No time/date column is given, so Years of Service is # used as the time-progression # proxy (x-axis). plt.figure(figsize=(7, 5)) plt.plot(df['YearsOfService'], df['PerformanceScore'], marker='o', color='steelblue') plt.title("Employee Performance Trend (by Years of Service)") plt.xlabel("Years of Service"); plt.ylabel("Performance Score") plt.tight_layout() plt.show() # 2. Bar chart - employee distribution across departments dept_counts = df['Department'].value_counts() plt.figure(figsize=(6, 5)) dept_counts.plot(kind='bar', color='darkorange') plt.title("Employee Distribution Across Departments") plt.xlabel("Department"); plt.ylabel("Number of Employees") plt.tight_layout() plt.show() # 3. Table - performance data per employee fig, ax = plt.subplots(figsize=(7, 2)) ax.axis('off') tbl = ax.table(cellText=df.values, colLabels=df.columns, loc='center', cellLoc='center') tbl.auto_set_font_size(False) tbl.set_fontsize(11) tbl.scale(1, 1.8) plt.title("Employee Performance Data", pad=20) plt.tight_layout() plt.show()</pre>	<pre>library(ggplot2) library(gridExtra) library(grid) df <- data.frame(EmployeeID = 1:3, Department = c('Sales', 'HR', 'Marketing'), YearsOfService = c(5,3,7), PerformanceScore = c(85,92,78)) # 1. Line chart - performance trend over time p1 <- ggplot(df, aes(x=YearsOfService, y=PerformanceScore)) + geom_line(color="steelblue") + geom_point(color="steelblue", size=3) + labs(title="Employee Performance Trend (by Years of Service)", x="Years of Service", y="Performance Score") + theme_minimal() print(p1) # 2. Bar chart - employee distribution across departments dept_counts <- as.data.frame(table(df\$Department)) colnames(dept_counts) <- c('Department', 'Count') p2 <- ggplot(dept_counts, aes(x=Department, y=Count)) + geom_col(fill="darkorange") + labs(title="Employee Distribution Across Departments", y="Number of Employees") + theme_minimal() print(p2) # 3. Table - performance data per employee grid.newpage() grid.table(df, rows=NULL)</pre>																
Output	Output																
Employee Performance Trend (by Years of Service)  <table border="1"><thead><tr><th>Years of Service</th><th>Performance Score</th></tr></thead><tbody><tr><td>3</td><td>92</td></tr><tr><td>5</td><td>85</td></tr><tr><td>7</td><td>78</td></tr></tbody></table>	Years of Service	Performance Score	3	92	5	85	7	78	Employee Performance Trend (by Years of Service)  <table border="1"><thead><tr><th>Years of Service</th><th>Performance Score</th></tr></thead><tbody><tr><td>3</td><td>92</td></tr><tr><td>5</td><td>85</td></tr><tr><td>7</td><td>78</td></tr></tbody></table>	Years of Service	Performance Score	3	92	5	85	7	78
Years of Service	Performance Score																
3	92																
5	85																
7	78																
Years of Service	Performance Score																
3	92																
5	85																
7	78																



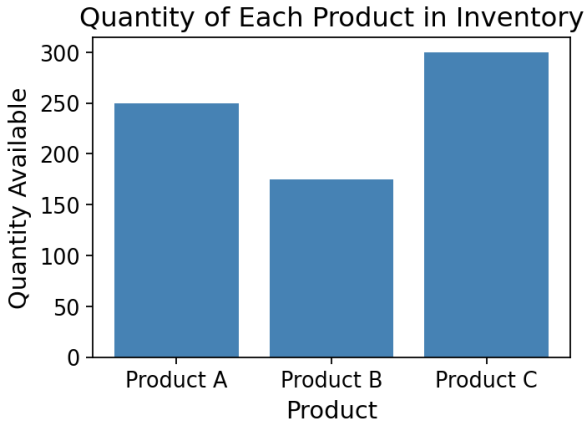
Employee Performance Data

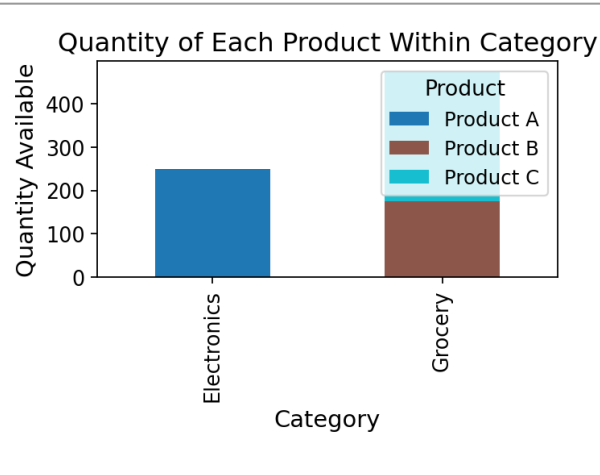
EmployeeID	Department	YearsOfService	PerformanceScore
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78

EmployeeID	Department	YearsOfService	PerformanceScore
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78

Experiment 18

Product Inventory Management

Python	R																
<pre>import matplotlib.pyplot as plt import pandas as pd df = pd.DataFrame({ 'ProductID': [1, 2, 3], 'ProductName': ['Product A', 'Product B', 'Product C'], 'QuantityAvailable': [250, 175, 300], 'Price': [20, 15, 18] }) # Category column is not in the given table, so it is assumed here to # satisfy Task 2 (stacked bar by category). df['Category'] = ['Electronics', 'Grocery', 'Grocery'] # 1. Bar chart - quantity of each product plt.figure(figsize=(7, 5)) plt.bar(df['ProductName'], df['QuantityAvailable'], color='steelblue') plt.title("Quantity of Each Product in Inventory") plt.xlabel("Product"); plt.ylabel("Quantity Available") plt.tight_layout() plt.show() # 2. Stacked bar chart - quantity by product within category pivot = df.pivot_table(index='Category', columns='ProductName', values='QuantityAvailable', fill_value=0) pivot.plot(kind='bar', stacked=True, figsize=(7, 5), colormap='tab10') plt.title("Quantity of Each Product Within Category") plt.xlabel("Category"); plt.ylabel("Quantity Available") plt.legend(title="Product") plt.tight_layout() plt.show() # 3. Table - inventory data per product fig, ax = plt.subplots(figsize=(7, 2)) ax.axis('off') display_df = df[['ProductID', 'ProductName', 'QuantityAvailable', 'Price']] tbl = ax.table(cellText=display_df.values, colLabels=display_df.columns, loc='center', cellLoc='center') tbl.auto_set_font_size(False) tbl.set_fontsize(11) tbl.scale(1, 1.8) plt.title("Product Inventory Data", pad=20) plt.tight_layout() plt.show()</pre>	<pre>library(ggplot2) library(gridExtra) library(grid) df <- data.frame(ProductID = 1:3, ProductName = c('Product A', 'Product B', 'Product C'), QuantityAvailable = c(250, 175, 300), Price = c(20, 15, 18)) # Category column is not in the given table, so it is assumed here to # satisfy Task 2 (stacked bar by category). df\$Category <- c('Electronics', 'Grocery', 'Grocery') # 1. Bar chart - quantity of each product p1 <- ggplot(df, aes(x=ProductName, y=QuantityAvailable)) + geom_col(fill="steelblue") + labs(title="Quantity of Each Product in Inventory", x="Product", y="Quantity Available") + theme_minimal() print(p1) # 2. Stacked bar chart - quantity by product within category p2 <- ggplot(df, aes(x=Category, y=QuantityAvailable, fill=ProductName)) + geom_col() + labs(title="Quantity of Each Product Within Category", y="Quantity Available") + theme_minimal() print(p2) # 3. Table - inventory data per product grid.newpage() grid.table(df[, c('ProductID', 'ProductName', 'QuantityAvailable', 'Price')], rows=NULL)</pre>																
Output	Output																
 Quantity of Each Product in Inventory <table border="1"><thead><tr><th>Product</th><th>Quantity Available</th></tr></thead><tbody><tr><td>Product A</td><td>250</td></tr><tr><td>Product B</td><td>175</td></tr><tr><td>Product C</td><td>300</td></tr></tbody></table>	Product	Quantity Available	Product A	250	Product B	175	Product C	300	 Quantity of Each Product in Inventory <table border="1"><thead><tr><th>Product</th><th>Quantity Available</th></tr></thead><tbody><tr><td>Product A</td><td>250</td></tr><tr><td>Product B</td><td>175</td></tr><tr><td>Product C</td><td>300</td></tr></tbody></table>	Product	Quantity Available	Product A	250	Product B	175	Product C	300
Product	Quantity Available																
Product A	250																
Product B	175																
Product C	300																
Product	Quantity Available																
Product A	250																
Product B	175																
Product C	300																



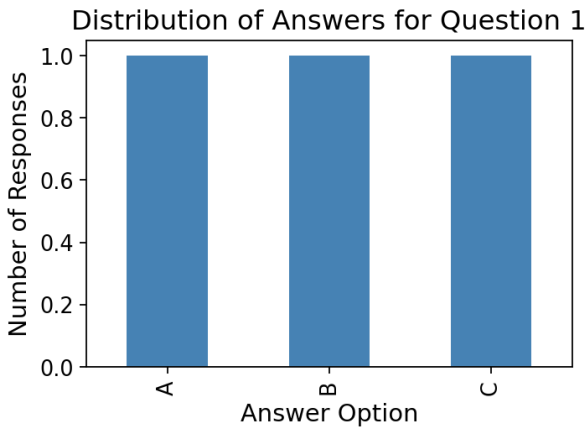
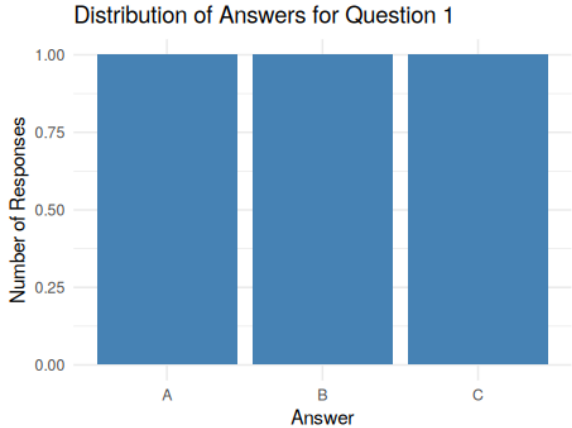
Product Inventory Data

ProductID	ProductName	QuantityAvailable	Price
1	Product A	250	20
2	Product B	175	15
3	Product C	300	18

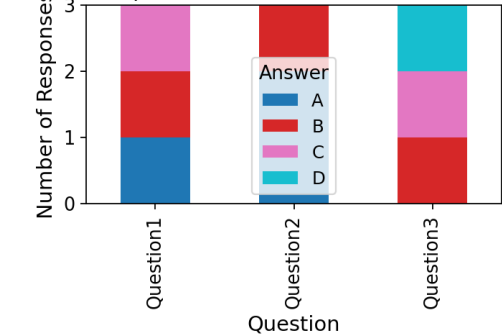
ProductID	ProductName	QuantityAvailable	Price
1	Product A	250	20
2	Product B	175	15
3	Product C	300	18

Experiment 19

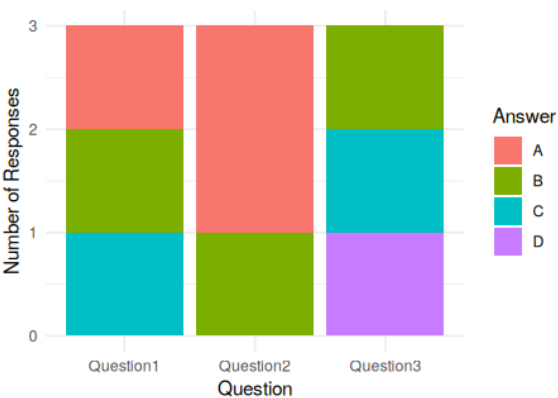
Survey Responses Analysis

Python	R
<pre>import matplotlib.pyplot as plt import pandas as pd df = pd.DataFrame({ 'SurveyID': [1, 2, 3], 'Question1': ['A', 'B', 'C'], 'Question2': ['B', 'A', 'A'], 'Question3': ['C', 'D', 'B'] }) # 1. Grouped bar chart - distribution of answers for # Question 1 q1_counts = df['Question1'].value_counts().sort_index() plt.figure(figsize=(6, 5)) q1_counts.plot(kind='bar', color='steelblue') plt.title("Distribution of Answers for Question 1") plt.xlabel("Answer Option"); plt.ylabel("Number of Responses") plt.tight_layout() plt.show() # 2. Stacked bar chart - overall distribution across # all three questions melted = df.melt(id_vars='SurveyID', value_vars=['Question1', 'Question2', 'Question3'], var_name='Question', value_name='Answer') counts = melted.groupby(['Question', 'Answer']).size().unstack(fill_value=0) counts.plot(kind='bar', stacked=True, figsize=(7, 5), colormap='tab10') plt.title("Overall Response Distribution Across All Questions") plt.xlabel("Question"); plt.ylabel("Number of Responses") plt.legend(title="Answer") plt.tight_layout() plt.show() # 3. Table - survey response data per survey fig, ax = plt.subplots(figsize=(6, 2)) ax.axis('off') tbl = ax.table(cellText=df.values, colLabels=df.columns, loc='center', cellLoc='center') tbl.auto_set_font_size(False) tbl.set_fontsize(11) tbl.scale(1, 1.8) plt.title("Survey Response Data", pad=20) plt.tight_layout() plt.show()</pre>	<pre>library(ggplot2) library(dplyr) library(reshape2) library(gridExtra) library(grid) df &lt;- data.frame(SurveyID = 1:3, Question1 = c('A', 'B', 'C'), Question2 = c('B', 'A', 'A'), Question3 = c('C', 'D', 'B')) # 1. Grouped bar chart - distribution of answers for # Question 1 q1_counts &lt;- as.data.frame(table(df\$Question1)) colnames(q1_counts) &lt;- c('Answer', 'Count') p1 &lt;- ggplot(q1_counts, aes(x=Answer, y=Count)) + geom_col(fill="steelblue") + labs(title="Distribution of Answers for Question 1", y="Number of Responses") + theme_minimal() print(p1) # 2. Stacked bar chart - overall distribution across # all three questions melted &lt;- melt(df, id.vars='SurveyID', variable.name='Question', value.name='Answer') counts &lt;- melted %>% count(Question, Answer) p2 &lt;- ggplot(counts, aes(x=Question, y=n, fill=Answer)) + geom_col() + labs(title="Overall Response Distribution Across All Questions", y="Number of Responses") + theme_minimal() print(p2) # 3. Table - survey response data per survey grid.newpage() grid.table(df, rows=NULL)</pre>
Output	Output
	

Overall Response Distribution Across All Questions



Overall Response Distribution Across All Questions



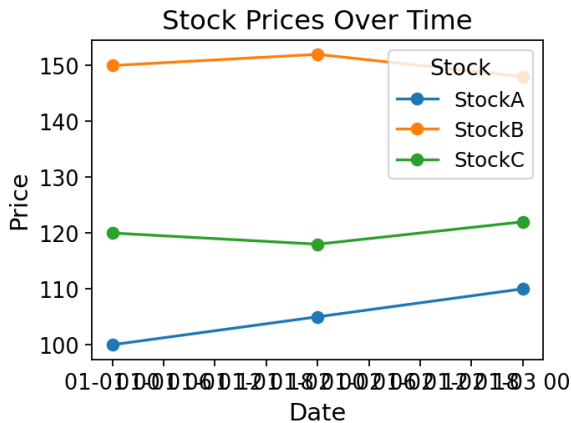
Survey Response Data

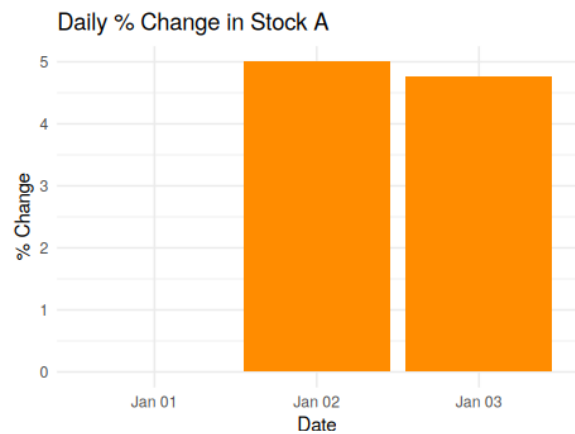
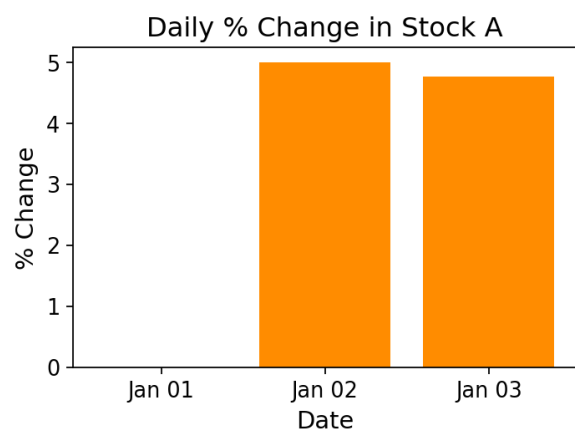
SurveyID	Question1	Question2	Question3
1	A	B	C
2	B	A	D
3	C	A	B

SurveyID	Question1	Question2	Question3
1	A	B	C
2	B	A	D
3	C	A	B

Experiment 20

Stock Analysis

Python	R
<pre>import matplotlib.pyplot as plt import pandas as pd df = pd.DataFrame({ 'Date': pd.to_datetime(['2023-01-01', '2023-01-02', '2023-01-03']), 'StockA': [100, 105, 110], 'StockB': [150, 152, 148], 'StockC': [120, 118, 122] }) # 1. Line chart - stock prices of three companies over time plt.figure(figsize=(7, 5)) for stock in ['StockA', 'StockB', 'StockC']: plt.plot(df['Date'], df[stock], marker='o', label=stock) plt.title("Stock Prices Over Time") plt.xlabel("Date"); plt.ylabel("Price") plt.legend(title="Stock") plt.tight_layout() plt.show() # 2. Bar chart - daily percentage change in Stock A df['StockA_PctChange'] = df['StockA'].pct_change() * 100 plt.figure(figsize=(6, 5)) plt.bar(df['Date'].dt.strftime('%b %d'), df['StockA_PctChange'].fillna(0), color='darkorange') plt.title("Daily % Change in Stock A") plt.xlabel("Date"); plt.ylabel("% Change") plt.tight_layout() plt.show() # 3. Table - stock price data for each company fig, ax = plt.subplots(figsize=(7, 2)) ax.axis('off') display_df = df[['Date', 'StockA', 'StockB', 'StockC']].copy() display_df['Date'] = display_df['Date'].dt.strftime('%Y-%m-%d') tbl = ax.table(cellText=display_df.values, colLabels=display_df.columns, loc='center', cellLoc='center') tbl.auto_set_font_size(False) tbl.set_fontsize(11) tbl.scale(1, 1.8) plt.title("Stock Price Data", pad=20) plt.tight_layout() plt.show()</pre>	<pre>library(ggplot2) library(reshape2) library(gridExtra) library(grid) df <- data.frame(Date = as.Date(c('2023-01-01', '2023-01-02', '2023-01-03')) , StockA = c(100,105,110), StockB = c(150,152,148), StockC = c(120,118,122)) # 1. Line chart - stock prices of three companies over time long_df <- melt(df, id.vars='Date', variable.name='Stock', value.name='Price') p1 <- ggplot(long_df, aes(x=Date, y=Price, color=Stock)) + geom_line() + geom_point() + labs(title="Stock Prices Over Time") + theme_minimal() print(p1) # 2. Bar chart - daily percentage change in Stock A df\$StockA_PctChange <- c(NA, diff(df\$StockA) / head(df\$StockA, -1) * 100) df\$StockA_PctChange[is.na(df\$StockA_PctChange)] <- 0 p2 <- ggplot(df, aes(x=format(Date, "%b %d"), y=StockA_PctChange)) + geom_col(fill="darkorange") + labs(title="Daily % Change in Stock A", x="Date", y="% Change") + theme_minimal() print(p2) # 3. Table - stock price data for each company grid.newpage() grid.table(df[, c('Date', 'StockA', 'StockB', 'StockC')], rows=NULL)</pre>
Output	Output
 The Python output shows a line plot titled "Stock Prices Over Time". The x-axis is labeled "Date" and ranges from 01-01 to 03-01. The y-axis is labeled "Price" and ranges from 100 to 150. Three lines are plotted: StockA (blue), StockB (orange), and StockC (green). StockB starts at 150 and ends at 148. StockC starts at 120 and ends at 122. StockA starts at 100 and ends at 110. A legend in the top right corner identifies the lines.	 The R output shows a line plot titled "Stock Prices Over Time". The x-axis is labeled "Date" and ranges from Jan 01 to Jan 03. The y-axis is labeled "Price" and ranges from 100 to 150. Three lines are plotted: StockA (red), StockB (green), and StockC (blue). StockB starts at 150 and ends at 148. StockC starts at 120 and ends at 122. StockA starts at 100 and ends at 110. A legend in the top right corner identifies the lines.



Stock Price Data

Date	StockA	StockB	StockC
2023-01-01	100	150	120
2023-01-02	105	152	118
2023-01-03	110	148	122

Date	StockA	StockB	StockC
2023-01-01	100	150	120
2023-01-02	105	152	118
2023-01-03	110	148	122